Ly-α Forest
LLS
Sub-DLA
Small-DLA
Large-DLA
TNG50-2 (High Res)
TNG50-4
TNG50-3
Impact Parameter b $[R_{200}]$
1.977
1.474
1.099
0.819
0.611
0.455
0.339
0.253
0.189
0.141
0.105
0.078
0.058
16
17
18
19
20
21
22
23
$\log N_{\rm HI}$ $[{\rm cm}^{-2}]$